

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct 19291369

I V.C. Bhawan Baisar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.C. Bhawan Baisar

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

- 1) Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, manageability, then justify the better choice.

Comparison:

FEATURE	TYPE-1 - HYPERVISOR	TYPE-2 - HYPERVISOR
Installation	Runs directly on physical hardware.	Runs on a host operating system.
Performance	High	Moderate
Security	More secure	Less secure
Manageability	Centralized Management	Managed through host OS
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation
Usage	cloud Data centers	Personal computers, testing.

Analysis of Performance, Security and Manageability:

Performance: Type - 1 hypervisors communicate directly with the hardware, resulting in faster execution, lower latency, and better resource utilization. Type - 2 hypervisors depend on the host operating system, which increases overhead and reduces performance.

Security: Type - 1 hypervisors provide better security because there is no host operating system between hardware and Virtual Machines. Type - 2 hypervisors have a larger attack surface since malware affecting host OS can also impact Virtual Machines.

Manageability: Type - 1 hypervisors support centralized management, live migration, load balancing and high availability making them suitable for large cloud environments. Type - 2 - hypervisors are easier to install but mainly suitable for testing and desktop Virtualization.

Justification:

Mission critical applications such as banking, healthcare, and government services require high availability, strong security, fast performance, and continuous operation.

Type - 1 - hypervisors meet these requirements by providing direct hardware access, efficient resource management, better VM isolation and enterprise level reliability.

Type - 2 - hypervisors are not suitable because they introduce additional overhead and depend on the host operating system.

2) A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how Virtualization architecture supports isolation and identify two major attack surfaces.

1) Virtualization Architecture for Workload Isolation:

Virtualization allows multiple Virtual Machines (VMs) to run independently on the same physical server using a hypervisor. Each VM has its own operating system, CPU, memory, storage and applications. This ensures that the finance, healthcare and student portal workloads remain isolated even though they share the same hardware. A failure or crash in one VM does not affect the other VMs.

2) How Virtualization supports Isolation:

- Each department runs in a separate Virtual Machine
- Every VM has its own operating system and dedicated virtual resources
- The hypervisor isolates CPU, memory, storage and network access between VMs.
- Unauthorized access between VMs is prevented.
- If one VM is infected by malware, the other VMs continue to operate securely.

- Resource allocation can be controlled independently for each workload.

3) Major Attack Surfaces:

a) Hypervisor Attack:

- Attackers target the hypervisor to gain control over multiple Virtual Machines.
- A compromised hypervisor can affect all hosted workloads.
- Protection: Regular updates, strong authentication and secure config.

b) VM Escape Attack:

- A malicious program inside one VM attempts to access the hypervisor or other VMs.
- This may expose sensitive data from finance or healthcare applications.
- Protection: Strong VM isolation, security patches and access control policies.

4) Recommended Solution:

The cloud provider should use a Type-1-hypervisor because it provides better isolation, security and performance. Separate VMs should be created for finance, healthcare, and student portals with dedicated virtual resources. Security measures such as encryption, firewalls, access controls, regular patching and continuous monitoring should also be implemented to protect against hypervisor and VM escape attacks. This ensures secure workload isolation while efficiently utilizing the same physical hardware.

3) Study the following VM host utilization snapshot and recommend corrective actions. CPU 92%; memory 88%, storage I/O latency 35ms, average VM boot time 170s. Explain the likely bottlenecks.

1) Analysis of the VM Host utilization:

- CPU utilization: = 92%. The processor is almost fully utilized, indicating that Virtual machines are competing for CPU resources.
- Memory utilization = 88%. Most of the RAM is occupied, leaving very little free memory for new or existing Virtual machines.
- Storage I/O Latency = 35ms. High storage latency means read and write operations are slow, reducing overall system performance.
- Average VM Boot time = 170 seconds. Virtual machines take a long time to start; indicating delays in accessing CPU, memory or storage resources.

2) Likely Bottlenecks:

CPU Bottleneck:

- CPU utilization of 92% indicates processor overload.
- Too many VMs are sharing limited CPU resources, causing slower application.

Memory Bottleneck:

- Memory utilization of 88% leaves little free RAM.
- The hypervisor may start memory swapping, which slows down VM.

Storage Bottleneck:

- Storage I/O latency of 35 ms is higher than normal.
- Slow disk access delays VM startup and application response.

VM Boot delay:

- The boot time 170 seconds is caused by the combined effect of high CPU usage, insufficient memory, and slow storage performance.

3) Recommended Corrective Actions:

- Increase CPU resources or migrate some VMs to another host using live migration.
- Upgrade RAM or allocate memory more efficiently to reduce swapping.
- Replace HDDs with SSDs or use faster storage systems to reduce swapping.
- Balance workload across multiple physical servers using load balancing.

4) Expected outcome:

- CPU utilization will decrease.
- Memory availability will improve.
- Storage latency will reduce, leading to faster read/write operations.
- VM boot will decrease significantly.
- Overall cloud performance, availability, and cost efficiency will improve, ensuring efficient utilization of the virtualized environment.

4) Analyse how software Defined Datacenter (SDC) architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage and network virtualization considerations.

1) SDC v/s Traditional Datacenter.

- A Traditional Datacenter uses physical servers, storage devices and networking equipment that must be configured manually. Resource allocation is slower and requires hardware installation.
- A Software Defined Datacenter (SDC) virtualizes compute storage and networking resources. All Infrastructure managed through software, making resource provisioning faster, flexible and automated.

Feature	Traditional Datacenter	Software Defined Datacenter (SDC)
Resource Management	Manual	Automated
Scalability	Limited	Highly Scalable
Deployment time	Hours or Days	Minutes
Flexibility	Low	High

2) Compute, Storage and Network Virtualization:

Compute:

- Physical servers are divided into multiple Virtual Machines (VMs)
- Resources such as CPU and Memory are allocated dynamically.
- Improves hardware utilization and allows quick deployment of applications.

Storage Virtualization:

- Combines multiple physical storage devices into a single virtual storage pool.
- Simplifies storage management and improves data availability.
- Allows user to access storage without knowing the physical location.

Network Virtualization:

- Creates virtual networks independent of physical hardware.
- Enables rapid network configuration and secure workload.
- Supports software-defined networking (SDN) for centralized management.

How SDC improves agility:

- Resources can be provisioned automatically within minutes.
- Virtual Machines can be created, migrated or deleted without changing physical hardware.
- Applications can scale up or down according to demand.
- Centralized software management reduces administrative effort.
- Faster deployment improves business continuity and service availability.

Practical Example:

A company launches an online shopping website during a festival sale. In a traditional data center, additional servers & storage are installed manually, causing delays. In SDDC, new VM, storage, virtual networks are created automatically through software. Allowing quick handling of increased customer traffic while maintaining high performance & availability. SDDC provides greater agility, flexibility, efficient resource utilization than datacenter.

- 5) A multi-cloud federation project fails due to identity mismatches between providers. Analyse the issue and propose a secure identity and privacy design using federation concepts.
- 1) Analysis of identity mismatch issue:

In a multi-cloud federation, organizations use services from multiple cloud providers such as AWS, Microsoft Azure, and Google Cloud. Each provider may have its own identity management system, user database, authentication method. These identity mismatches reduce security and make user management difficult.

- 2) Federation concepts for secure identity.

Identity federation allows users to access multiple cloud services using a single digital identity.

Key Federation concepts include:

- Single Sign-on (SSO): Users log in once and access multiple cloud services.
- Identity Provider (IdP): Authenticates the user and provides Identity Information.
- Service Provider (SP): Accepts the authenticated Identity & grants access.
- Federation Protocols: Standards such as SAML, OAuth 2.0, and OpenID Connect (OIDC) securely exchange authentication information between providers.

3. Source Identity and Privacy Design:

- Use a centralized Identity Provider (IdP) for user authentication.
- Implement Single Sign-On (SSO) to avoid multiple user accounts.
- Use SAML, OAuth 2.0 or OpenID Connect for source identity federation.
- Enable Multi Factor Authentication (MFA) for stronger security.
- Encrypt user data during storage and transmission.
- Apply Role-Based Access Control (RBAC) so users can access only authorized resources.
- Regularly monitor user activities and maintain audit logs.